DeepMind

Deep Reinforcement Learning with Plasticity Injection

Evgenii Nikishin, Junhyuk Oh, Georg Ostrovski, Clare Lyle, Razvan Pascanu, Will Dabney, Andre Barreto

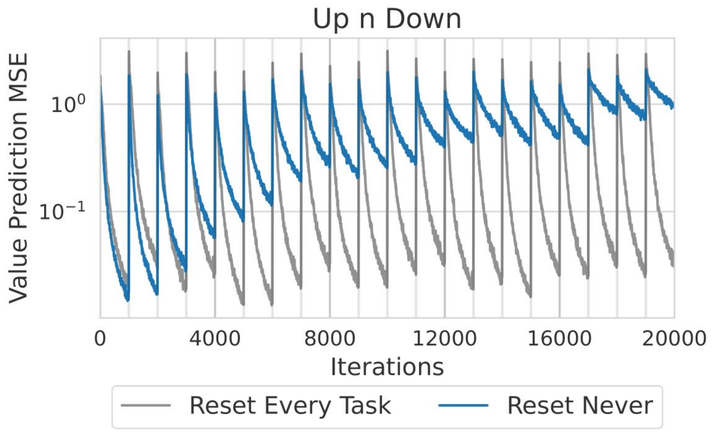
Open Review .net

Overview

- Deep RL agents **lose plasticity** during training, but detecting it is challenging
- Plasticity Injection is a tool for careful **analysis** of the phenomenon
- Plasticity Injection **reincarnates** agents that lost plasticity without the need to re-train
- Agents with Plasticity Injection **save compute** by dynamically growing networks

Background on Plasticity Loss

On a sequence of policy evaluation tasks, a network becomes worse at predicting values



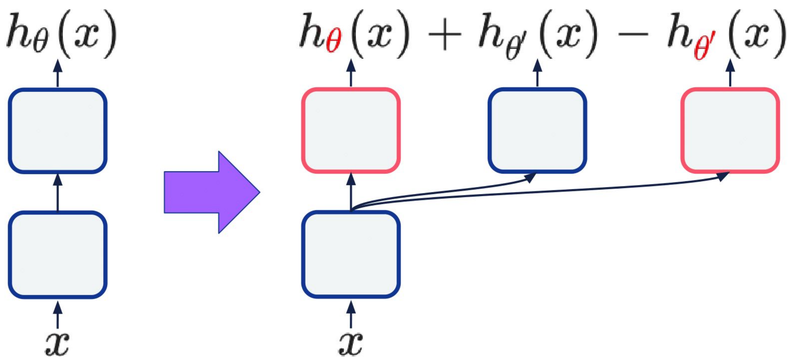
Challenges in Deep RL

- No clear measure of network plasticity
- Performance in deep RL can be bottlenecked by many factors including exploration

How to tell if an agent cannot improve due to loss of plasticity?

Plasticity Injection

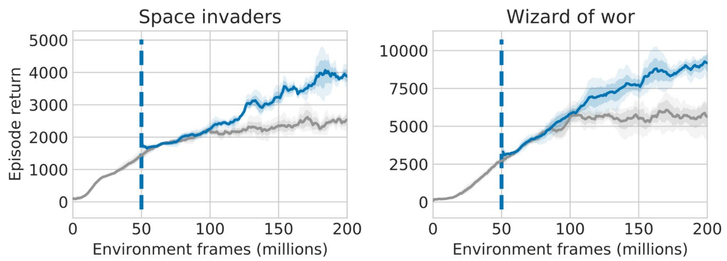
At some point, freeze the network and create a new, learning a residual to predictions



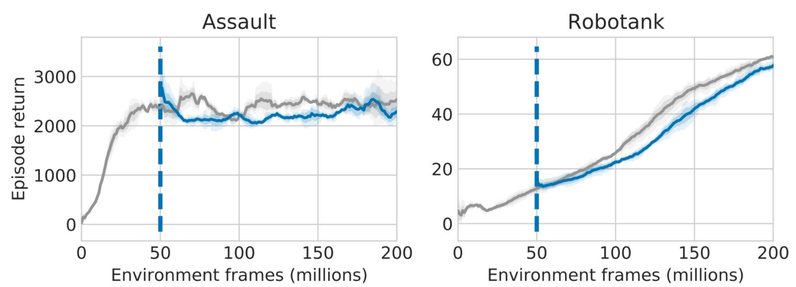
Key properties:

- Same number of trainable parameters
  → roughly **keeps representational power**
- Unaffected predictions after intervening
  → **excludes exploration** confounders

Diagnosing Loss of Plasticity

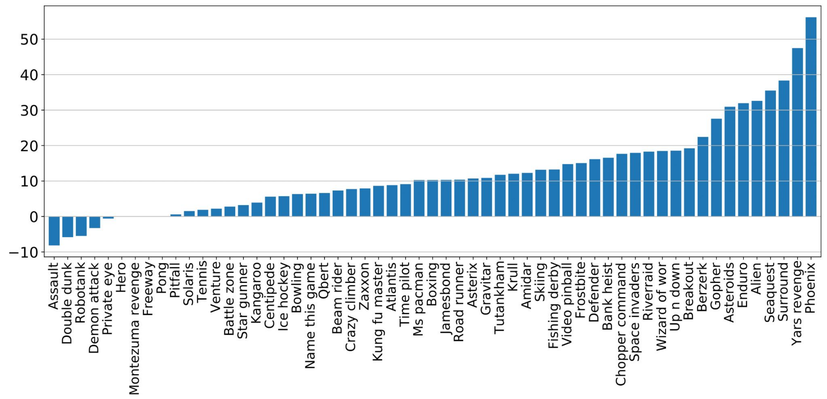


Improves performance → agent lost plasticity



Agent is stuck due to exploration / learns well → no effect from plasticity injection

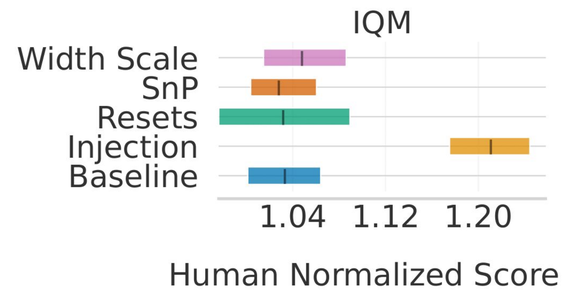
Results on 57 Atari Games



Agents lose plasticity within their lifetime

Reincarnating with Injection

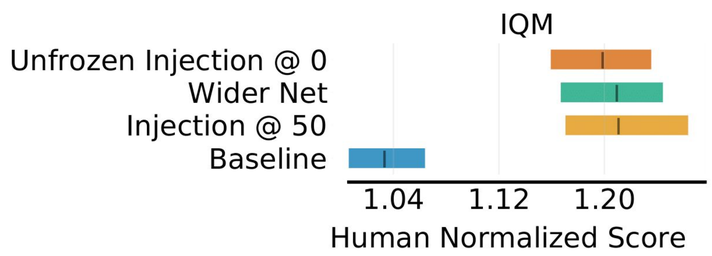
Keep learning after plasticity loss efficiently



Better than re-training-free alternatives

Dynamic Network Growth

Do we need all plasticity from the beginning?



Same IQM as large nets from start but cheaper